Who Patches the Lightning Network?

Release Lines from Gossip, Node Platforms and Security Advisories, 2023–2026

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***Abstract*—Every Lightning implementation disclosed security vulnerabilities in 2026, and every advisory told operators to upgrade. Whether they do has never been measured, because Lightning gossip carries no software version. We show that the release line of a public node can be read from the feature bits it announces: matched against the source of every release of LND, Core Lightning, Eclair and LDK, the most likely line is exact for 132 of 136 node–line pairs of nodes that state their version. Applied to 814 daily snapshots of the public graph from 2023 to 2026, it shows that the network patches on release and platform schedules, not on disclosures. Half of the LND nodes on an older line upgrade within 88 days of a release, while small nodes and nodes that skipped earlier releases lag far behind. The largest movements are node platforms shipping a release to all their users: the three largest upgrade waves came about a day after the Umbrel app store released a new LND version, and since June 2025, 17% of LND upgrades fell in the 4.4% of the time just after a platform release. Disclosure, a median 256 days after the fix, reached only operators already close to current: of 12 disclosures we can test, only Core Lightning's of July 2026 raised the rate at which nodes left affected releases, 3.87-fold among nodes one release line behind, while nodes further behind moved when BTCPay Server's own advisory had their operators update. No LND disclosure produced a detectable increase; two months after Lightning Labs' August 2026 batch, 71% of the nodes exposed at disclosure still ran an affected release. The lever on that tail is the platforms that package releases.**

***Index Terms*—Lightning Network, Bitcoin, patch deployment, vulnerability disclosure, node platforms, software versions, measurement, gossip**

I. Introduction

Between July and October 2026, every major Lightning implementation disclosed security vulnerabilities. Lightning Labs published ten LND advisories on a single day [1] and eight more a month later [2]; Core Lightning (CLN) asked operators who could not upgrade to restart their nodes offline [3]; Eclair and LDK shipped security releases. Each advisory ends with the same advice: upgrade. Whether operators do, and how fast, decides how long a disclosed vulnerability stays exploitable. For web servers, browsers and Bitcoin and Ethereum full nodes, this question has been measured for two decades [4]–[8]. For the Lightning Network it has not, because Lightning gossip carries no software version: earlier studies could tell which implementation a node runs [9]–[11], not which release.

We show that the release can be read from gossip after all, and use it to ask who patches the Lightning Network: operators on their own schedule, the platforms that package the implementations, or the advisories themselves. Each release announces a characteristic set of protocol feature bits. From the source of every release of LND, CLN, Eclair and LDK since 2019 we derive which sets each release can produce, and match them against 814 daily snapshots of the public graph taken by our own node from May 2023 to October 2026. The answer is that the network patches on release and platform schedules, not on disclosures. Most operators upgrade within months of a release; the largest movements are platforms shipping a release to all their users at once; and an advisory, published long after its fix, moves at most the operators who were already close to current. We make four contributions.

- **Release-line inference from gossip.** We infer the release line (the finest unit gossip separates; patch releases are invisible) of every public node with a known implementation. On 124 CLN nodes whose alias states their version, the inferred range contains the true line on 99.99% of 15,549 node-days and the most likely line is exact for 132 of 136 node–line pairs. For LND, which has no such ground truth, three checks that need none agree: line decreases are 0.7% of observed line changes, no LND line released since 2024 was seen on more than 70 nodes before its publication, and 97–99.7% of the nodes that changed line right after a platform shipped a new LND line landed on the line it shipped (§IV).
- **Upgrade dynamics.** With a managed fleet of up to 1,781 live LND nodes [11] set apart, new lines reach half of live LND nodes in 71–79 days. Half of the LND nodes on an older line upgrade within 88 days of a release, but the smallest quarter of nodes takes 135, and a node already more than one line behind has about five times lower odds of moving in any week (§V).
- **Platforms set the pace.** We date the LND and CLN releases of two node platforms to the minute from public records. The three largest LND upgrade waves, 262–385 nodes each, re-signed their announcements a median 16–27 hours after the Umbrel app store published a new LND version, and another wave followed BTCPay Server's security release. Upgrades signed within 48 hours after such a release are 17% of all LND upgrades since June 2025, although those windows cover 4.4% of the

time (§VI).

- **Disclosure, response and exposure.** For the 30 LND and CLN advisories whose fix boundary gossip can see, we measure how many live nodes and how much capacity remained exposed. Fixes typically shipped months before disclosure (median 256 days). For 12 disclosures we compare the weekly rate at which exposed nodes moved to a later line before and after disclosure, with and without the moves that platform releases explain. Only CLN's July 2026 disclosures raised it, by a factor of 3.87 among nodes one line behind; nodes further behind moved when BTCPay's own advisory had their operators update. No LND disclosure produced a detectable increase, and 71% of the nodes exposed to Lightning Labs' August 2026 batch were still live on an affected release two months later (§VII).

All data behind the figures and tables, and the code, are released. We report aggregates only: no list, ranking or label of individual nodes by release, platform or exposure.

II. Background and Related Work

A. Gossip and feature bits

Public Lightning nodes announce themselves and their channels through gossip [12]. A node's `node_announcement` carries its alias, colour, network addresses and a feature vector: a bit field in which each pair of bits marks a protocol feature as required (even bit) or optional (odd bit) [13]. Implementations set their feature vector from compiled-in defaults, which change when a release adds or retires a feature, plus whatever options the operator switches on. Gossip carries no version string. A node re-signs its announcement when it restarts or changes its settings, so a node that upgrades and restarts re-announces with the feature vector of the new release.

B. Node platforms

Many public nodes are not installed by hand. Node-in-a-box distributions such as Umbrel, Start9 and RaspiBlitz, merchant stacks such as BTCPay Server, and hosted services package an implementation and decide when their users receive a new release. Umbrel's app store is a public git repository: a device sees a new version of its Lightning app once the change to the app's manifest is merged [14], which dates each release to the minute. Gossip does not name the platform a node runs on.

C. Inferring implementations

Zabka et al. classify public nodes into LND, c-lightning and Eclair from default channel parameters, colours and aliases [9], [15]. Their study contains the only prior observation of an LND upgrade we found: after LND changed its default CLTV delta from 144 to 40 in March 2019, the new value overtook the old one around mid-December, about nine months later. They name the planning of critical software updates as a use of implementation inference but do not measure it. Espinasa-Vilarrasa et al. infer the implementation from gossip traffic or from announced feature bits with decision trees [10]. Kappos et al. note that their gossip snapshots carried no version information, and fell back on owner-reported fields for 370 nodes [16]. Our previous paper used feature-bit rules to measure client concentration and the 2026 Core Lightning shutdown [11]. All of these stop at the implementation; none infers which release a node runs.

Longitudinal studies of the Lightning graph cover topology, centralisation and robustness [17]–[21], infrastructure [22], privacy [23], channel lifecycles [24], gossip convergence [25] and, most recently, stability and node lifetimes [26]–[28]. None of them studies software versions. When Lightning Labs disclosed CVE-2019-12999, they were "surprised to find many nodes running very old versions" but published no numbers [29].

D. Versions in other blockchain networks

Bitcoin and Ethereum peers report their software version in the handshake, so crawlers can count versions directly. Kim et al. found 68% of Geth nodes two or more releases behind and tracked client age against release dates [8]. Neudecker reports that Bitcoin nodes adopt a release mostly in its first two months, and documents an alternative client that went from zero to more than 700 peers in a day [30]; Park et al. and Eisenbarth et al. survey Bitcoin node versions and their known vulnerabilities [31], [32]. SRLabs estimated that about 40% of Parity nodes were still vulnerable a month after an advisory [33]. Zcash sidesteps slow upgrades by expiring old releases at network upgrades [34]. The Lightning Network offers neither a version string to count nor a forced expiry.

E. Patch deployment

The window of vulnerability, from disclosure to patch, was framed by Arbaugh et al. [4]. Rescorla probed servers affected by the OpenSSL vulnerabilities of 2002 and found more than two thirds still vulnerable two weeks after disclosure [5]. Frei et al. showed that automatic updates dominate how fast a client population patches [6], and Sarabi et al. that the vendor's patch speed matters more than users' promptness [35]. Durumeric et al. followed Heartbleed across the Internet and found that direct notification raised patching by about half [7]; Li et al. found that most notified operators never act [36]. Nappa et al. modelled patch deployment with survival analysis [37], and Li and Paxson measured the life cycle of security fixes from the code side [38]. We bring this lens to a network where versions must first be inferred, and where a release is often fixed months or years before the vulnerability is disclosed.

III. Data and Vantage Point

a) Gossip snapshots: We take a daily dump of the public graph (`describegraph`) from our own LND node and keep, per day, every node announcement (feature bits, alias, colour, addresses, last-update time) and every channel direction (the policy a node sets for forwarding towards its peer, with capacity, the *disabled* flag and last-update time). An announcement's last-update time is the moment its node signed it, so the first announcement that shows a new release's

bits dates an upgrade more finely than the daily snapshot does (§VI). The snapshots run from May 2023 to the data cutoff of 13 October 2026. A per-day quality check discards dumps that are partial (fewer than half the channels of their period), stale (fewer than half the channels updated within two weeks) or error responses written while the node's wallet was locked; 814 days remain. Daily coverage is dense from November 2023, with gaps in May–June 2024 (partial dumps), August 2024 and January–May 2025 (wallet locked after restarts). Upgrade times observed across a gap are known only to lie inside it; we treat them as interval-censored (§V).

b) Vantage: All snapshots are one node's view. LND prunes channels that have not been updated for about two weeks, and our view widened in July 2024 (from about 8.4k to 11–13k nodes with channels), after the node's own peer set grew. We compare levels within a period and report shares, which are less sensitive to the view; in particular, the few live nodes on releases from before 2021 appear only from July 2024. Only announced (public) nodes are visible: mobile wallets and services behind unannounced channels, where much of LDK and Eclair runs, are not.

c) Who is still there: Nodes that stop running do not leave the graph at once: their peers mark the channels towards them disabled, and the channels stay announced until they are closed or pruned. An offline node cannot announce anything, so liveness must be read from its peers' side [11]. We call a node *live* on a day when at least one channel direction towards it, set by a peer, is enabled and was updated in the previous 14 days; its *live capacity* is the capacity of those channels. Results with 7- and 28-day windows are in Table V. The distinction matters. On the cutoff day, 2,689 LND nodes with channels (outside the managed fleet described next) announced a release older than 0.19, but only 434 of them were live; counting every announced node would put the share of LND nodes on those releases at 42.3%, against 12.5% of live nodes.

d) A managed fleet: Our previous paper found a managed fleet of small LND nodes with aliases of the form `LNT.<name>`, one or two channels each, mostly to a handful of hubs with the same prefix, and counted up to 1,980 announced fleet nodes [11]. We use its definition (the alias prefix, or at least half of a node's channels to a fleet hub) but count live nodes. The fleet appeared on 20 June 2025 and peaked on 7 December 2025 at 1,781 live nodes, 28% of all live LND nodes, with 1.3 BTC of live capacity against 8,300 BTC for live LND as a whole; at the cutoff 566 of its nodes were still live (14% of live LND nodes). Its nodes share one Lightning Terminal feature set and upgrade in step (§VI), so they are not independent operators, and we exclude them from every per-node statistic.

IV. Inferring Release Lines from Gossip

A. Release lines

We built a table of the `node_announcement` feature bits of every release of the four implementations from their source at each release tag: LND 0.7–0.21.4, Core Lightning (CLN) 0.8.2–26.06.9, Eclair 0.3–0.14.3 and LDK 0.0.113–0.2.7. For each release we record the bits it sets by default, the bits an operator can add through options, flags, plugins or known custom extensions, and the default bits an operator can switch off; each entry cites the source file it comes from. A release's date is its publication date (GitHub release page; crates.io for LDK), the day operators could install it. Consecutive releases with identical defaults form a *release line*, the finest unit gossip can resolve: LND has 10 lines from 0.8 to 0.21, CLN 13, Eclair 13 and LDK 6. Some lines are single minor releases (LND 0.18, 0.19, 0.20, 0.21); others span several (LND 0.15.1–0.17; LDK 0.0.120–0.1 covers January 2024 to December 2025). Patch releases never change the defaults, so a patch level is never visible.

For a node with feature set B, a line ℓ with defaults D_ℓ, optional bits O_ℓ and removable bits R_ℓ is *consistent* when $D_\ell \setminus R_\ell \subseteq B \subseteq D_\ell \cup O_\ell$. The *strict range* is every consistent line released by the day of observation; it contains the running line under any non-developer configuration. The *likely line* is the consistent line that needs the fewest non-default bits among the lines released by the day the node's current feature set first appeared. Dating the inference to that day matters: a node cannot run a line that did not exist when it began announcing its bits, and without it a new release would relabel every node whose existing bits happen to fit the new line (233 CLN nodes running 24.05 with the onion-message option would have "upgraded" to 24.08 on its release day). A set consistent only with lines not yet released is a pre-release build; a set consistent with no line is unrecognised.

B. Implementation

Release lines are defined per implementation, so the implementation comes first. We extend the feature-bit rules of our previous paper [11]: LND-only bits (2023, 31) and Eclair-only bits (37, or 29 with 61) are decisive; a set with bits 11 and 39 or 43 is CLN when bit 5 is set and Eclair otherwise; 39, 43 or 63 without 11 is LDK. New in this paper, the release table itself is evidence. When a set is consistent with the lines of exactly one implementation, that implementation is assigned, which recovers releases older than the distinguishing bits, such as LND 0.8–0.11 and CLN before 23.05; and when a bit rule names an implementation that has no release producing the set while exactly one other implementation does, the table wins. Weak signals (LND's default colour or alias, default CLTV deltas) label the remaining nodes with low confidence, and only operator-attributed and high-confidence labels receive a release line. Announcements that are not visible on a day (last update 0) carry the node's previous feature set forward.

C. Validation

a) Implementation: 797 nodes name their implementation in their alias ("…[LND]", "…-CLN", "eclair", "ldk-node"); LND's default alias is excluded because the rules use it. On the latest day each was announced with channels, the labels reach precision 100.0% and recall 96.4% for LND (664 nodes) and precision 98.3% and recall 99.1% for CLN (117

nodes) (Table I). The misses are almost all low-confidence labels, which receive no release line.

b) Release line: CLN built in developer mode puts the version into the default alias (e.g. `YELLOWWATER-v25.09`); the alias follows the binary, so it is correct on every day the node is announced. Over 15,549 announced node-days of 124 such nodes on 11 lines (no operator above 5% of node-days), the strict range contains the stated line on 99.99% of node-days and the likely line is exact on 98.2% (Fig. 1). Node-days of one node are not independent, so the node is the better unit: of the 136 combinations of node and stated line, 132 are exact on every day (97%, 95% CI 93–99%). The misses are a 24.05 node with the onion-message option, which makes it identical to 24.08; a 25.02.1 node and a 23.11.1 node whose switched-off options make them look one line older; and a single day of a release candidate.

c) Checks without ground truth: No LND node in our data states its version in its alias, so for LND, where most of our dynamics are measured, we rely on checks that a wrong release table would fail. First, operators rarely return to an older release, while a noisy inference would make nodes flicker between lines: among consecutive observations of live nodes, the likely line decreases in 0.7% of line changes for LND (against 13,802 increases) and in 0.8% for CLN. Second, a line assigned the wrong bits would show nodes on it long before its release: each LND line released since 2024 was seen on at most 70 nodes before its publication, against 2,628–3,756 in its first 90 days. Third, a check from outside gossip: a platform that ships a new LND version moves its nodes to a known line (§VI). Of the 1,165 line changes whose new announcement was signed within 48 hours after one of the 4 platform releases that introduced a new LND line, 97–99.7% landed on the line that release shipped.

d) What gossip cannot resolve: Among live nodes with a release line, the strict range spans more than one line on 11% of LND and 47% of CLN node-days in 2026, because options let an older line imitate a newer one; the likely line is unique on all of them. Unrecognised sets are 1.0% (LND) and 0.1% (CLN) of node-days, and pre-release builds about 0.1% of LND node-days. Three limits remain. Patch releases are invisible. A node that upgrades without restarting, or whose new announcement does not reach us, keeps its old bits. Developer overrides and forks can imitate any set. The first two bias exposure upward, the third in either direction; none can be measured from gossip alone, which is why we report the strict and likely ranges side by side.

V. How Lightning Nodes Upgrade

A. Adoption of new release lines

Fig. 2 shows the release-line mix of live LND and CLN nodes. Each new LND line displaces its predecessor within a few months but leaves a tail: at the cutoff, 12.5% of live LND nodes were still on a line older than 0.19, released in May 2025. New lines reached half of live LND nodes 71–79 days after release (Fig. 3); half of live capacity moved as fast or faster for 0.18 and 0.19 (41 and 67 days) but more slowly

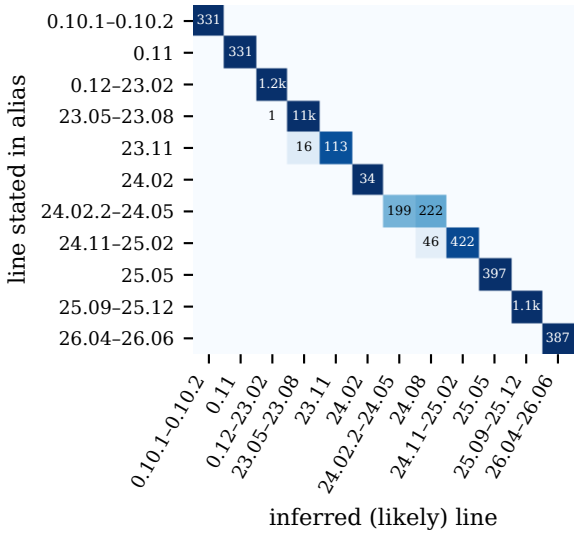


Fig. 1. Release line stated in CLN developer aliases (rows) against the likely line inferred from feature bits (columns), in node-days. Shading is the row share.

TABLE I
Validation against self-identifying aliases. Top: client labels of 797 nodes whose alias names their implementation. Bottom: release lines of CLN nodes whose alias states the running version, graded on every announced node-day.

Client	Nodes	Precision (%)	Recall (%)
LND	664	100.0	96.4
CLN	117	98.3	99.1
Eclair	9	58.3	77.8
LDK	7	100.0	100.0

CLN nodes / lines with a stated version	124 / 11
Node-days graded	15,549
Strict range contains the stated line (%)	99.99
Likely line equals the stated line (%)	98.2
Node–line pairs exact on every day	132 / 136
95% CI (%)	[93, 99]

for 0.20 (118 days). Counting the fleet as independent nodes would put LND 0.20 at 50% of nodes after 45 days instead of 79. CLN lines took 66–216 days. Every LND line from 0.18 on was announced by a few nodes 53–176 days before its release: developers and early adopters running unreleased builds on mainnet.

B. Time to upgrade

Adoption curves mix upgrades with nodes arriving and leaving. To follow operators, we take every node that is live on an older line on the release day of LND 0.19, 0.20 or 0.21 or of a CLN line from 23.11 on, and measure the time until it is first seen on the new line or a later one. Snapshot gaps leave the upgrade day inside an interval (for 87% of LND upgrades it is at most two days wide); nodes that stop being live before upgrading are censored at their last live day. The share upgraded by day t is the nonparametric maximum-

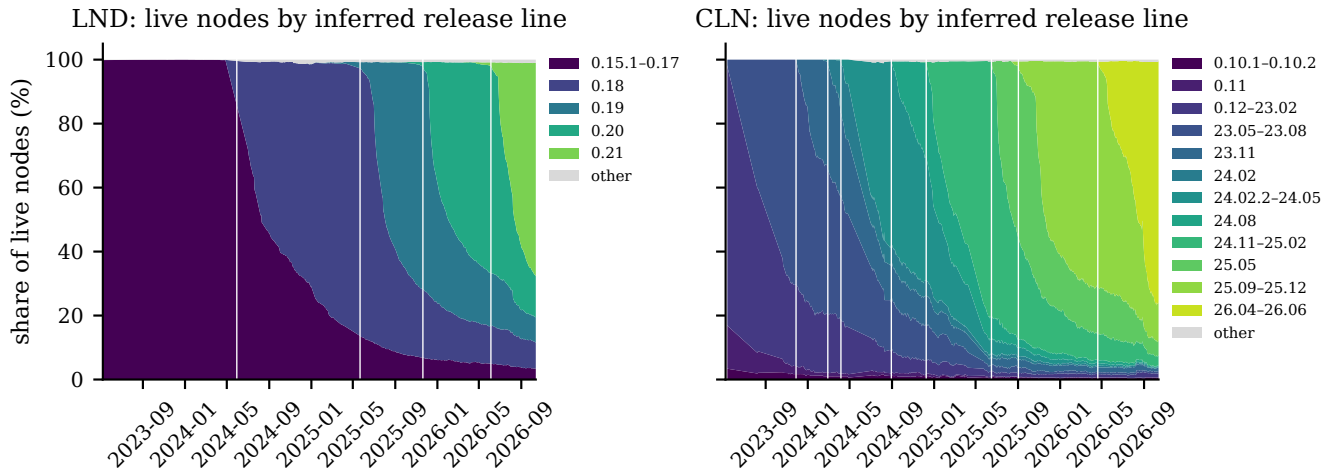


Fig. 2. Live LND and CLN nodes by inferred release line (likely line), excluding the managed fleet. White lines mark releases. Lines under 2% of nodes throughout are grouped as "other".

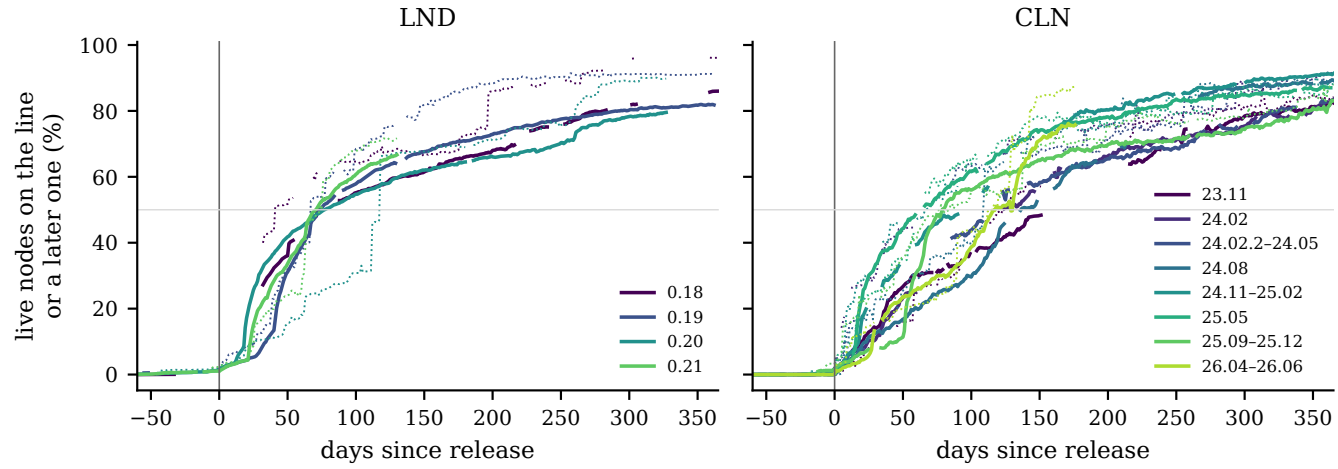


Fig. 3. Share of live nodes (solid) and of their live capacity (dotted) on each release line or a later one, by days since the line's release. Lines are broken across snapshot gaps longer than three days.

likelihood estimate for interval-censored data [39]. Half of the LND nodes upgraded within 88 days and 76% within a year; for CLN the median is 137 days. Size matters at the bottom: the smallest quarter of LND nodes by live capacity took a median 135 days, the largest quarter 75 (Fig. 4).

Table II fits a discrete-time model of the weekly probability of upgrading for LND. Larger nodes upgrade sooner (odds ratio 1.11 per tenfold capacity, 1.14 per tenfold channel count), and so do Tor-only nodes (1.61), which make up most of the platform waves of §VI. A node more than one line behind when the release ships is far less likely to move (0.19): the tail is made of nodes that skipped releases before. Nodes with LND's default colour are no slower (1.08), and nodes seen before 2024 upgrade at the same rate as newer ones (1.01).

TABLE II
WHAT PREDICTS UPGRADING: DISCRETE-TIME LOGIT ON NODE-WEEKS AFTER EACH LND RELEASE (WEEK AND RELEASE FIXED EFFECTS; STANDARD ERRORS CLUSTERED BY NODE). ODDS RATIOS WITH 95% CI.

Covariate	Odds ratio	95% CI
Capacity (×10)	1.11	[1.07, 1.16]
Channels (×10)	1.14	[1.06, 1.24]
Tor-only	1.61	[1.52, 1.71]
LND default colour	1.08	[1.03, 1.14]
Seen before 2024	1.01	[0.96, 1.07]
>1 line behind	0.19	[0.18, 0.21]
Nodes / node-weeks	7,239 / 220,525	

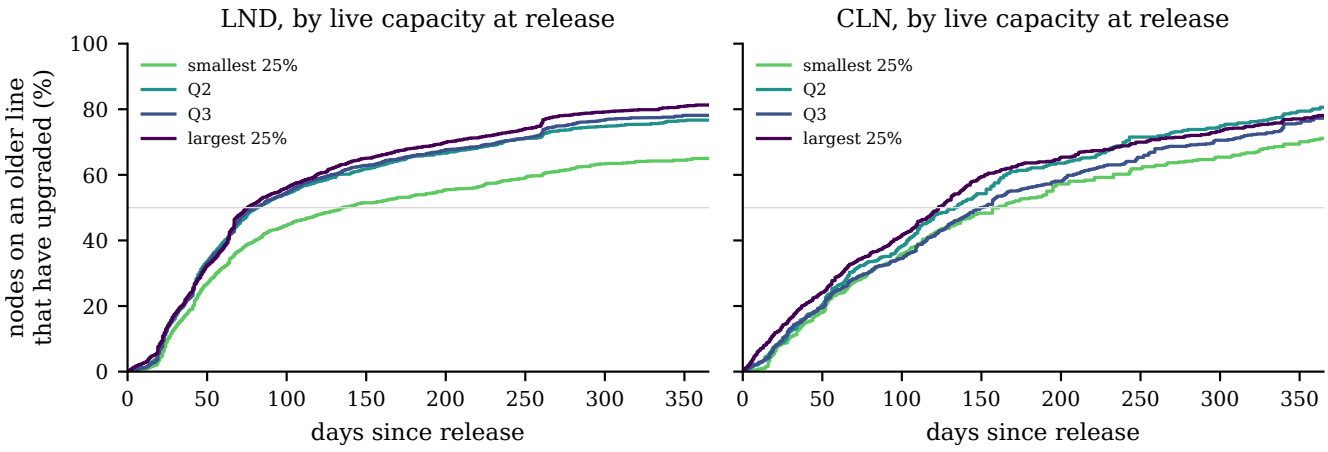


Fig. 4. Share of nodes that were live on an older line at a release and have since been seen on the new line or a later one, by quartile of live capacity at the release (Turnbull estimate; LND 0.19–0.21, CLN 23.11–26.04).

VI. Platforms Set the Pace

A. Dating platform releases

Gossip does not name the platform a node runs on, but the signing time of a node's first announcement on a new line can be set against platform releases dated to the minute. For Umbrel we take the merge time of every pull request that changed the version of its LND or CLN app from 2024 to the cutoff [14]. A BTCPay Server deployment updates only when its operator runs the update, which installs the LND or CLN version BTCPay pins at that moment, so its routine version bumps reach nodes over weeks; we take its security release 2.4.2 of 7 August 2026, which asked every operator to update at once [40]. Start9, RaspiBlitz, myNode and hosted services publish through channels we cannot date to the day and are not covered.

B. Upgrade waves

Upgrades arrive in bursts (Fig. 5). We call a day a wave day for a line when more nodes moved to it than on 99% of all (day, line) pairs, and merge consecutive wave days on one line into an event. Outside the fleet, LND had 6 events of 114–385 nodes, 19–67 days after the line's release. They are not gossip catching up after a gap: in every event, 100% of the nodes had signed their new announcement within two days before we first saw it.

The three largest LND events, 262–385 nodes each, follow Umbrel's releases of LND 0.19.1, 0.20.0 and 0.21.1: in each, at least 87% of the nodes signed their new announcement within 48 hours after the merge (median 16–27 hours after) and at most 1% in the 24 hours before. These nodes are far more often Tor-only (84–92%, against 61–73% of all live LND nodes), small (median live capacity 0.045–0.050 BTC, against 0.055–0.056 BTC for the other upgraders to the same line) and established (median 366–617 days in our data). Their timing is the platform's: each wave came 19–41 days after LND's own release, when Umbrel shipped it. The managed fleet of §III kept the same schedule: 327 of its nodes moved to LND 0.20 on 9 December 2025, 100% of them within 48 hours after Umbrel published 0.20.0, with the same median lag (26 hours) as the platform's other users.

A fourth event is different: 114 nodes on 8 August 2026, 99% of them signed within 48 hours after BTCPay's security release (median 21 hours). They are mostly clearnet (26% Tor-only, against 60%), as merchant nodes are, and 48 of them came from lines older than 0.20, skipping at least one release. Their operators had not updated for months and did so when the advisory for their merchant software told them to; that advisory concerned BTCPay itself, not LND. Of the remaining two events, one (160 nodes, 28 July 2025) was signed before Umbrel merged LND 0.19.2 (86% in the 24 hours before), so that release cannot explain it, and one (204 nodes, 20 July 2024) matches no release we can date. CLN shows the same pattern at its smaller scale: its one attributed event, 27 nodes on 18 May 2026, followed Umbrel's release of CLN 26.04.1 (93% within 48 hours, median 6 hours).

C. Beyond the waves

Platform releases account for a share of upgrades out of proportion to the time they cover. Since June 2025, 17% of LND upgrades whose day is known to within two days were signed within 48 hours after a platform release that ships the new line, although these windows cover 4.4% of the time; for CLN, 14% against 5.4%. Since the platforms we cannot date are left out, these shares are lower bounds. The pushes also check the inference (§IV-C): of the 1,165 line changes signed within 48 hours after the 4 platform releases that introduced a new LND line, 97–99.7% landed on the line that release shipped.

VII. Did Disclosure Move Operators?

A. Advisories gossip can see

We collected every public security advisory for LND, CLN, Eclair and LDK from January 2022 to the cutoff (release

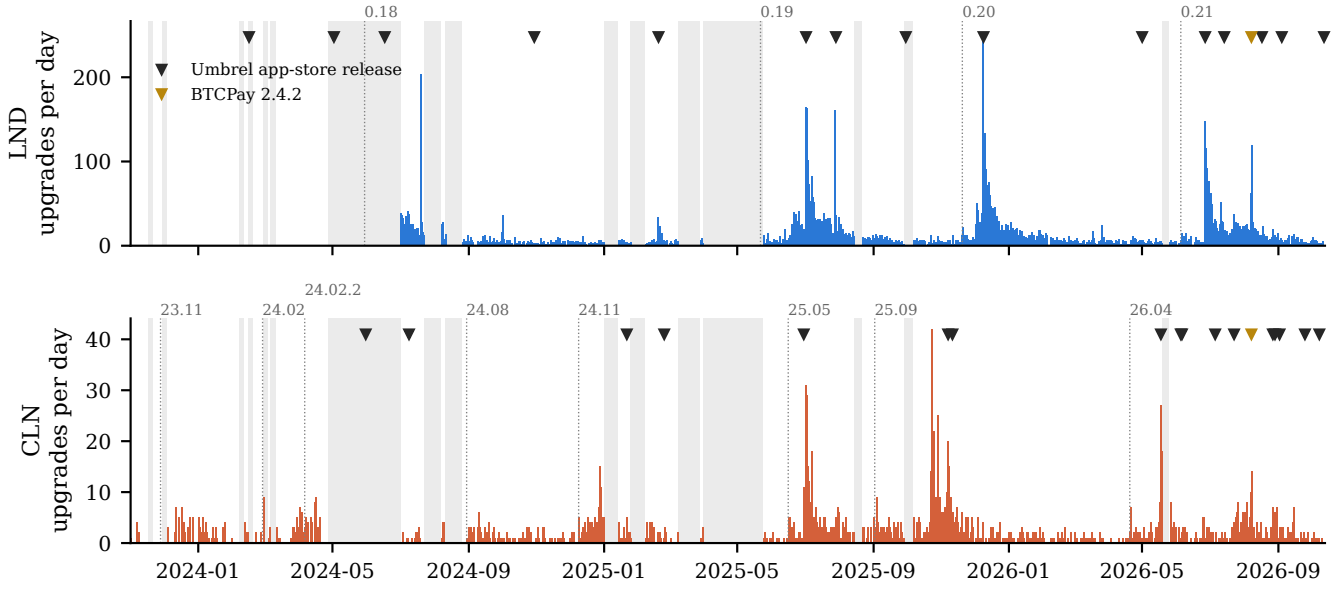


Fig. 5. Nodes first seen on a newer release line per day, where the upgrade day is known to within two days (managed fleet excluded). Grey: snapshot gaps; dotted lines: LND and CLN releases; triangles: Umbrel app-store releases (black) and BTCPay Server's security release 2.4.2 (gold) of that implementation.

notes, Lightning Labs' advisory index, Delving Bitcoin and authors' write-ups), recording the affected versions, the first fixed release and the disclosure date, each from the source cited in Table III. An advisory is visible in gossip when its fix boundary coincides with the start of a release line (LINE), or falls inside a line but leaves whole lines affected or fixed on either side (PARTIAL, which gives bounds). Fixes inside the newest line, such as CLN 26.06.7–26.06.9 and Eclair 0.14.1–0.14.3, are patch releases and invisible. Eclair and LDK have too few live public nodes with a known line (about ten Eclair nodes) for shares to be meaningful, so Table III lists the 23 LND and CLN entries, covering 30 advisories.

Two clocks run for each advisory. Fixes usually ship long before the vulnerability is disclosed: the median gap between the first fixed release and disclosure is 256 days, and the six advisories Lightning Labs published on 11 August 2026 concerned bugs fixed in LND 0.19.0, released 446 days earlier. A node still on an affected line at disclosure has therefore usually skipped a release that was already available for many months. Exposure is the share of live nodes (excluding the managed fleet) on an affected line, with the share of their live capacity alongside. It is largest where the fix is youngest: when Teinturier disclosed that LND accepted too few confirmations on cooperative closes, 69 days after the fix shipped in 0.21.0, 50.6% of live LND nodes, holding 52.5% of live LND capacity, were on an affected line.

B. Response to disclosure

If operators act on advisories, nodes should leave the affected lines faster after a disclosure than before it. For every advisory with whole lines affected and enough snapshots on both sides (at least six of the eight weeks before and three of the four weeks after), we take, at the start of each week, the live nodes on a wholly affected line, and count those first seen on a later line within the week. The rate is moves per node-week, and we compare the weeks before and after disclosure by their rate ratio. Moves are not independent: a platform push or a group of operators moves many nodes in the same week, and the weekly counts vary up to 70 times more than Poisson counts would. The 95% confidence intervals are therefore quasi-Poisson, scaled by that dispersion, and we add an exact permutation test of whether the weekly rates after disclosure exceed those before. Platform pushes could produce a jump that no operator chose, so we also report the ratio without moves signed within 48 hours after a platform release that ships the new line (§VI). The comparison has a known bias: the rate of leaving a line falls as the line ages and its most attentive operators leave, so the before-window has the higher baseline, most of all when a disclosure follows a release by only a few months. A ratio near one therefore rules out only a large response.

12 disclosures qualify, 9 for LND and 3 for CLN (Table IV, Fig. 6). Only CLN's two disclosures of 19 July 2026, memory-exhaustion vulnerabilities in gossipd (fixed in 26.04) and connectd (fixed in 25.09) [45], show a clear increase. Among live CLN nodes on lines before 26.04, the weekly rate rose from 1.6% to 6.2% (ratio 3.87, 95% CI 2.43–6.17), and for lines before 25.09 from 1.6% to 5.4% (3.45, 0.98–12.08). The two groups responded differently. The 334 nodes one line behind, on 25.09–25.12, moved steadily over the four weeks: without the moves that followed platform releases their ratio is 3.87 (2.33–6.46), and each of the four weeks after disclosure had a higher rate than any of the seven before (permutation $p = 0.003$). The 186 nodes on older lines moved

TABLE III

ADVISORIES WHOSE FIX BOUNDARY IS VISIBLE IN GOSSIP (LND AND CLN). EXPOSURE = SHARE OF LIVE NODES OF THAT IMPLEMENTATION WHOSE INFERRED RELEASE LINE IS AFFECTED, IN %; SHARE OF THEIR LIVE CAPACITY IN PARENTHESES; FOR PARTIAL ROWS (THE FIX LANDS INSIDE A LINE) THE BOUNDS ARE GIVEN IN BRACKETS AT DISCLOSURE. FIX→DISC. = DAYS FROM THE FIRST FIXED RELEASE TO PUBLIC DISCLOSURE. # = ADVISORIES GROUPED IN THE ROW.

Impl.	Advisory (disclosed)	Class	#	Fixed in	Fix→disc.	At disclosure	+60 d	Cutoff
CLN	Fake channel DoS [41] (2023-08-23)	PARTIAL	1	23.02	175	–	6 (2)	1 (0)
CLN	Replacement cycling (CVE-2023-40234) [42] (2023-10-16)	PARTIAL	1	23.08.1	34	38 [24–100] (17)	26 (10)	2 (0)
CLN	BOLT11 parsing bugs, batch 1 [43] (2023-12-08)	PARTIAL	1	23.02	282	4 [4–95] (2)	2 (1)	1 (0)
CLN	BOLT11 parsing bugs, batch 2 [43] (2023-12-08)	LINE	1	23.11	10	96 (96)	69 (73)	3 (0)
CLN	funding_created assertion DoS [44] (2026-05-16)	LINE	1	25.09	256	29 (24)	24 (21)	12 (2)
CLN	connectd queue memory exhaustion [45] (2026-07-19)	LINE	1	25.09	320	24 (20)	14 (3)	12 (2)
CLN	gossipd unknown-SCID memory exhaustion [45] (2026-07-19)	LINE	1	26.04	90	67 (58)	29 (16)	24 (13)
CLN	Ping-flood OOM [46] (2026-08-25)	LINE	1	25.09	357	19 (16)	–	12 (2)
LND	Fake channel DoS [41] (2023-08-23)	PARTIAL	1	0.16.0	147	–	0 (0)	1 (0)
LND	Replacement cycling (CVE-2023-40233) [42] (2023-10-16)	PARTIAL	1	0.17.0	13	0 [0–100] (0)	0 (0)	1 (0)
LND	Premature channel_update OOM [47] (2023-12-21)	LINE	1	0.14.0	764	0 (0)	0 (0)	1 (0)
LND	channel_update signature after rate limit [47] (2023-12-21)	PARTIAL	1	0.15.0	546	0 [0–0] (0)	0 (0)	1 (0)
LND	Onion bomb (CVE-2024-38359) [48] (2024-06-18)	PARTIAL	1	0.17.0	259	–	–	1 (0)
LND	Irrevocable fees / update_fee breach fee-burn [49] (2024-12-10)	PARTIAL	1	0.18.3	89	33 [33–100] (25)	24 (11)	4 (0)
LND	Excessive HTLC failback [50] (2025-03-04)	LINE	1	0.18.0	278	20 (8)	–	4 (0)
LND	gossip_timestamp_filter goroutine DoS [51] (2025-07-22)	PARTIAL	1	0.18.3	313	11 [11–60] (3)	9 (2)	4 (0)
LND	Replacement stalling, failback #2, infinite inbox [52]–[54] (2025-12-04)	LINE	3	0.19.0	196	28 (12)	22 (10)	13 (6)
LND	Zero-timestamp gossip panic [55] (2026-06-18)	PARTIAL	1	0.20.1	126	34 [34–97] (27)	25 (16)	20 (10)
LND	Lightning Labs batch: gossip, peer and queue DoS [1] (2026-08-11)	LINE	6	0.19.0	446	15 (6)	13 (6)	13 (6)
LND	HTLC first-stage sweep budget failure [56] (2026-08-11)	LINE	1	0.19.0	446	10 (5)	8 (6)	8 (6)
LND	Too few confirmations on cooperative close [57] (2026-08-13)	LINE	1	0.21.0	69	51 (53)	34 (29)	33 (28)
LND	Gossip stall via malformed announcements [58] (2026-09-21)	PARTIAL	1	0.20.1	221	8 [8–32] (5)	–	8 (4)
LND	Brontide write-allocation OOM [59] (2026-09-21)	LINE	1	0.20.0	305	22 (11)	–	20 (10)

Before July 2024 our view did not include the few live nodes on LND releases older than 0.14 (§III); exposures to the 2023 LND advisories at disclosure are lower bounds. In 2023 the mixed LND line 0.15.1–0.17 held almost every LND node, so the bounds of the 2023 LND PARTIAL rows span 0–100%. –: no usable snapshot within 7 days.

mostly in the 48 hours after BTCPay's security release of 7 August, which installed CLN 26.06.1 on every BTCPay deployment that updated; without those moves their ratio is 1.54 (0.47–5.09; $p = 0.212$). The operators who acted on CLN's disclosure were those already close to current; the ones further behind moved when their platform's own advisory had them update. A month later, the ping-flood vulnerability, fixed at the same 25.09 boundary, was disclosed [46], and the next day the maintainers asked operators to upgrade or restart offline [3]. The 125 nodes still on older lines may have moved somewhat faster, but not clearly: 1.48 (0.36–6.13), and outside platform releases, whose removal takes BTCPay's push out of the before-window, 1.99 (1.03–3.84; $p = 0.077$).

No LND disclosure produced a detectable increase. Across the 9, the ratios are at most 1.48, and none is significant (permutation $p \geq 0.12$, with or without platform moves). The intervals are wide, however. For Lightning Labs' batch of 11 August 2026, the weekly rate was 1.4% before and 1.4% after (0.99, 0.22–4.56); its before-window holds BTCPay's security release four days earlier, and without the moves that followed it the ratio is 1.30 (0.56–3.03). Outside platform releases, a modest increase is compatible with these data, but one as large as CLN's (3.87) is not. What the rates leave open, the exposed cohort settles (§VII-C): two months after the batch, only 10% of the nodes exposed at disclosure had upgraded. The lowest ratios belong to the cooperative-close disclosure, 69 days after 0.21.0, and the two of 21 September 2026, which followed the platform pushes of June and August: the bias described above.

C. What became of the exposed nodes

On the day Lightning Labs disclosed its batch, 14.9% of live LND nodes, holding 6.4% of live LND capacity, were on a line older than 0.19 (Fig. 7). Of the 550 exposed nodes, 54 had upgraded by the cutoff, 108 were no longer live and 388 (71%) were still live on an affected line. Morehouse's disclosure of three LND vulnerabilities fixed in the same release, on 4 December 2025 [52]–[54], found 27.7% of live LND nodes exposed; four months later 55% of that cohort was still live on an affected line, 18% had upgraded and 28% were gone. The exposed LND nodes are small: their share of capacity is about half their share of nodes.

CLN's exposure fell much further. Of the 524 CLN nodes exposed to the gossipd vulnerability at disclosure, 243 (46%) had upgraded by the cutoff and 170 (32%) were no longer live, leaving 111; exposure fell from 66.6% of live CLN nodes to 24.2% (from 57.8% to 12.7% of capacity). Much of the fall is nodes leaving rather than upgrading, as in the CLN shutdown we measured earlier [11]; leaving also ends the exposure, and restarting offline was what the maintainers asked of operators who could not upgrade.

D. Robustness

Table V varies the definitions behind the four case studies. A 7- or 28-day liveness window moves the exposure by about one percentage point. Counting the fleet as nodes would lower LND's exposure, since the fleet ran fixed lines, to 12.9% instead of 14.9% for Lightning Labs' batch. The strict lower bounds for LND sit well below the likely-line exposure, because a newer line with some default features switched off

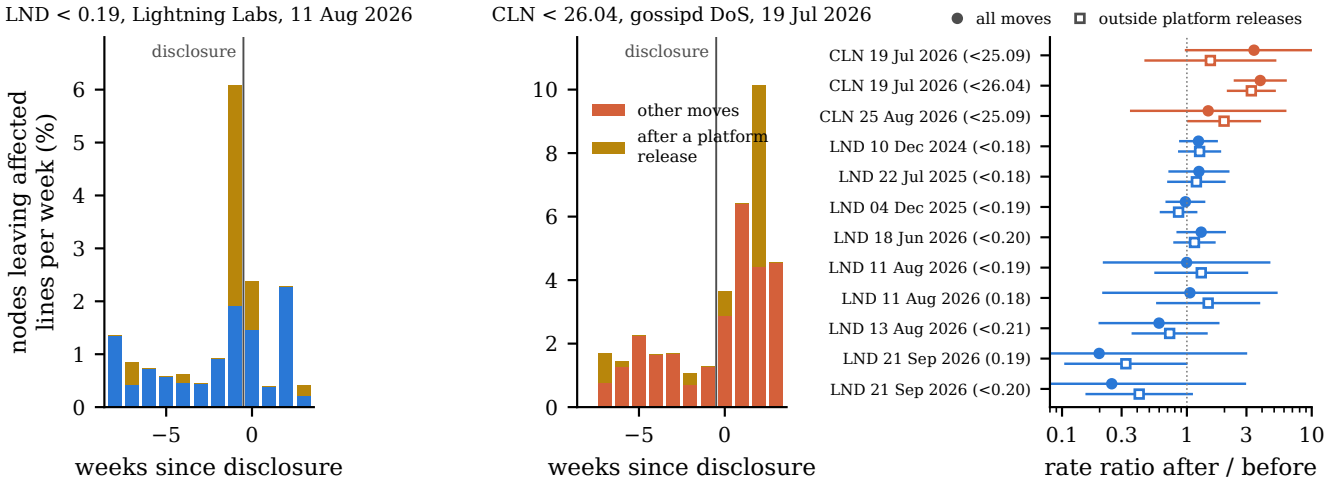


Fig. 6. Left and centre: weekly share of the live nodes on an affected line that moved to a later line, around two disclosures; gold: moves signed within 48 hours after a platform release that ships the new line. Right: rate ratio after/before disclosure with quasi-Poisson 95% CI for every disclosure that can be tested, for all moves (filled) and outside platform releases (open). Managed fleet excluded.

TABLE IV

DID OPERATORS RESPOND TO THE DISCLOSURE? WEEKLY SHARE OF LIVE NODES ON A WHOLLY AFFECTED LINE THAT MOVED TO A LATER LINE, IN THE 8 WEEKS BEFORE AND THE 4 WEEKS AFTER DISCLOSURE (FEWER WHERE SNAPSHOTS ARE MISSING OR THE CUTOFF INTERVENES), AND THE RATIO AFTER/BEFORE WITH A QUASI-POISSON 95% CI. "OUTSIDE PLATFORM RELEASES" LEAVES OUT MOVES SIGNED WITHIN 48 H AFTER AN UMBREL OR BTCPAY RELEASE THAT SHIPS THE NEW LINE (SECTION VI); p: EXACT ONE-SIDED PERMUTATION TEST THAT THE WEEKLY RATES AFTER EXCEED THOSE BEFORE, OUTSIDE PLATFORM RELEASES. MANAGED FLEET EXCLUDED.

		Fix→disc.	At risk	Weeks	Weekly rate (%)		Rate ratio after/before		
Impl.	Advisory (disclosed; affected lines)	(days)	nodes	bef./aft.	before	after	all moves	outside platform releases	p
CLN	connectd queue memory exhaustion (2026-07-19; <25.09)	320	186	7/4	1.6	5.4	3.45 [0.98, 12.08]	1.54 [0.47, 5.09]	0.212
CLN	gossipd unknown-SCID memory exhaustion (2026-07-19; <26.04)	90	520	7/4	1.6	6.2	3.87 [2.43, 6.17]	3.28 [2.14, 5.04]	0.003
CLN	Ping-flood OOM (2026-08-25; <25.09)	357	125	8/4	3.4	5.0	1.48 [0.36, 6.13]	1.99 [1.03, 3.84]	0.077
LND	Irrevocable fees / update_fee breach fee-burn (2024-12-10; <0.18)	89	1,861	8/3	1.2	1.4	1.24 [0.89, 1.73]	1.26 [0.87, 1.84]	0.133
LND	gossip_timestamp_filter goroutine DoS (2025-07-22; <0.18)	313	542	8/3	0.7	0.9	1.25 [0.73, 2.14]	1.19 [0.71, 2.00]	0.327
LND	Replacement stalling, failback #2, infinite inbox (2025-12-04; <0.19)	196	1,273	8/4	1.6	1.6	0.97 [0.69, 1.37]	0.86 [0.62, 1.19]	0.810
LND	Zero-timestamp gossip panic (2026-06-18; <0.20)	126	1,420	8/4	1.3	1.6	1.30 [0.84, 2.01]	1.15 [0.80, 1.66]	0.214
LND	Lightning Labs batch: gossip, peer and queue DoS (2026-08-11; <0.19)	446	546	8/4	1.4	1.4	0.99 [0.22, 4.56]	1.30 [0.56, 3.03]	0.295
LND	HTLC first-stage sweep budget failure (2026-08-11; 0.18)	446	363	8/4	1.7	1.8	1.06 [0.21, 5.21]	1.48 [0.58, 3.78]	0.232
LND	Too few confirmations on cooperative close (2026-08-13; <0.21)	69	1,821	8/4	6.5	3.9	0.60 [0.20, 1.78]	0.73 [0.37, 1.44]	0.891
LND	Gossip stall via malformed announcements (2026-09-21; 0.19)	221	292	8/3	5.3	1.1	0.20 [0.01, 2.97]	0.32 [0.11, 0.99]	0.976
LND	Brontide write-allocation OOM (2026-09-21; <0.20)	305	763	8/3	3.2	0.8	0.25 [0.02, 2.92]	0.41 [0.16, 1.09]	0.970

can produce the bits of an older one (§IV); the likely line assumes the fewest such changes. The exposed nodes are not stale entries in our view of the graph: 77% of the LND nodes exposed to Lightning Labs' batch had signed their current announcement, with the affected line's feature bits, in the 14 days before disclosure (against 95% of all live LND nodes). CLN nodes re-sign far less often (15% of all live CLN nodes within 14 days), so CLN exposure rests more on the absence of a new announcement.

VIII. DISCUSSION

a) Platforms, not advisories, move nodes: The largest upgrade events in our data are platform pushes. Each time the Umbrel app store shipped a new LND line, 262–385 nodes re-announced within about a day; BTCPay Server's security release moved 114 LND nodes in a day, and with them the CLN nodes on old lines that CLN's own disclosure had not moved. No LND disclosure produced a visible increase at all. Frei et al. found the same for browsers: the update mechanism, more than the user, sets the patch level [6]; and Sarabi et al. found the vendor's patch speed to matter more than users' promptness [35]. On the Lightning Network the vendor that reaches most small nodes is the platform, not the implementation. Each Umbrel wave came 19–41 days after LND's own release, a large part of the median time to upgrade (88 days). Two practical consequences follow. A fix reaches most small nodes when platforms ship it, so the platforms' release schedule, not only the implementation's, decides how long a vulnerability stays open. And a platform that tells its users to update moves even nodes that ignored earlier releases: 48 of the 114 nodes in the BTCPay wave had skipped at least one LND line, although the advisory that moved them concerned BTCPay rather than LND.

b) Disclosure reaches the operators who were already upgrading: The LND advisories of 2025 and 2026 were published long after the fixes shipped (median 256 days across

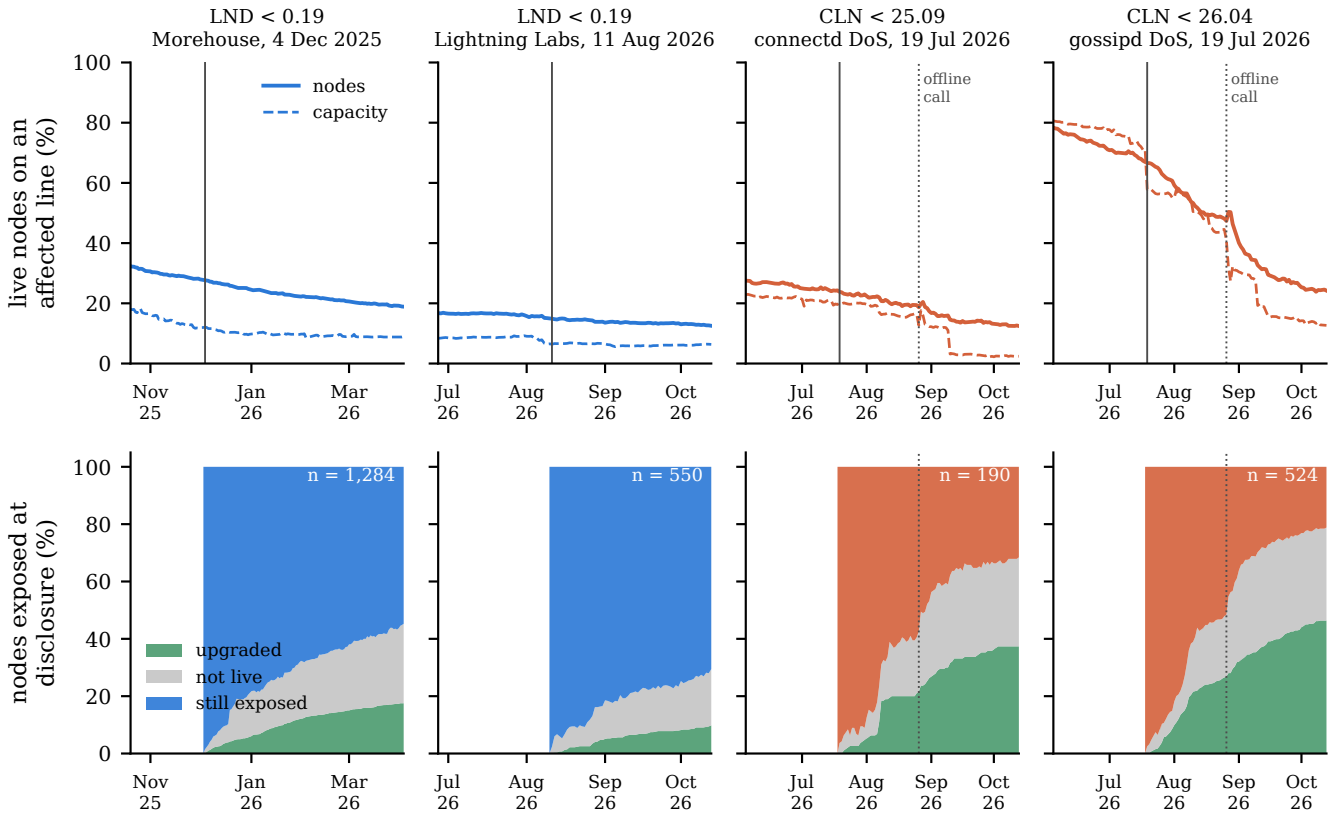


Fig. 7. Top: share of live nodes (solid) and of their live capacity (dashed) on a line affected by the advisory; solid vertical line: disclosure; dotted: CLN's call of 26 August 2026 to upgrade or restart offline. Bottom: the nodes exposed on the last snapshot before disclosure, by what became of them. Managed fleet excluded.

the LND and CLN advisories), a common practice that lets operators upgrade before attackers learn what to look for. It works for the operators who follow releases: half of LND nodes move to a new line within three months, and the largest move first. The nodes still on an affected line at disclosure are the ones that did not, and for LND the disclosure did not detectably change their pace: 71% of those exposed to Lightning Labs' August batch were still exposed two months later. The one disclosure that did move operators makes the same point from the other side. CLN's July disclosures raised the rate of leaving affected lines 3.87-fold among nodes one line behind, operators who keep up; the nodes further behind moved when BTCPay's advisory had their operators update. Reported attacks do not explain the CLN response, since the maintainers reported attacks on unpatched nodes only in October [60]. More plausibly, a fix that had shipped 90 days before disclosure found most exposed CLN nodes one line behind, while one shipped 446 days before found only the LND nodes that had skipped every release since. Notification studies on the Internet find that most notified operators never act [36] and that direct notification raised Heartbleed patching by about half [7]. A Lightning advisory is not a notification: it is a post that reaches those who read the implementation's channels, and the operators who do not are reached, if at all, by their platform. Release-line inference also gives maintainers a measurement they have lacked: the share of live nodes and capacity still on an affected line, read in aggregate from public gossip, which can inform when to disclose.

c) The tail: What remains exposed are small nodes, nodes that skipped earlier releases, and nodes that no platform updates. For advisories that require a channel with the victim, these nodes are also the cheapest to reach. Their share of capacity at risk is small (6.4% of live LND capacity for the August 2026 LND batch), but each holds its operator's funds.

d) Counting nodes: At its peak, one coordinated fleet was more than a quarter of the live LND nodes we observe, with almost no capacity. Any statistic that counts nodes, whether implementation shares, version shares or exposure, would have been dominated by that fleet's schedule. Fleets are cheap to create and hard to see. This one gave itself away by a shared alias and hubs [11] and, here, by upgrading in step; a fleet with neither would pass for hundreds of independent operators. Node-level Lightning measurements should look for such fleets before counting, and report capacity alongside nodes.

TABLE V

ROBUSTNESS OF THE CASE-STUDY EXPOSURES: SHARE OF LIVE NODES ON AN AFFECTED LINE (%) AT DISCLOSURE AND AT THE CUTOFF, WITH LIVE CAPACITY IN PARENTHESES FOR THE MAIN DEFINITION. LIVENESS: A PEER-SET CHANNEL DIRECTION TOWARD THE NODE ENABLED AND UPDATED WITHIN 7, 14 (MAIN) OR 28 DAYS. STRICT BOUNDS: NODES WHOSE EVERY CONSISTENT LINE IS AFFECTED, AND NODES WITH ANY AFFECTED CONSISTENT LINE. LAST COLUMN: SHARE OF THE EXPOSED NODES WHOSE CURRENT ANNOUNCEMENT WAS SIGNED IN THE PREVIOUS 14 DAYS, AGAINST 94–96% OF ALL LIVE LND AND 15–22% OF ALL LIVE CLN NODES ON THE SAME DAYS (SECTION VII-D).

Advisory	When	Day	14 d (main)	7 d	28 d	Strict bounds	Fleet as nodes	Signed <14 d
LND < 0.19, Morehouse, 4 Dec 2025	disclosure	2025-12-04	28 (12)	27	28	[7, 28]	21	89
	cutoff	2026-10-13	13 (6)	12	13	[4, 13]	11	74
LND < 0.19, Lightning Labs, 11 Aug 2026	disclosure	2026-08-11	15 (6)	14	15	[5, 15]	13	77
	cutoff	2026-10-13	13 (6)	12	13	[4, 13]	11	74
CLN < 25.09, connectd DoS, 19 Jul 2026	disclosure	2026-07-19	24 (20)	24	24	[24, 24]	–	12
	cutoff	2026-10-13	12 (2)	12	12	[12, 12]	–	15
CLN < 26.04, gossipd DoS, 19 Jul 2026	disclosure	2026-07-19	67 (58)	66	67	[63, 100]	–	10
	cutoff	2026-10-13	24 (13)	24	24	[21, 100]	–	14

IX. LIMITATIONS

Release lines, not patches. Gossip cannot separate patch releases, so fixes inside a line, including all of CLN's 2026 patch releases, cannot be measured; for advisories with a fix inside a line we report bounds. *Inference.* An operator can switch features on or off so that an older line looks like a newer one or the reverse; the strict range covers every non-developer configuration, and validation on self-identifying CLN nodes bounds the error of the likely line. For LND we have no ground truth at scale, only the three checks of §IV-C. A node that upgrades without re-announcing, or whose announcement we do not receive, keeps its old bits, which biases exposure up. *Platforms.* We can date the releases of two platforms; upgrades pushed by others count as operators' own, so platform shares are lower bounds. Dropping every move in the 48 hours after a platform release also drops operators who moved on their own in those hours, which makes the ratios outside platform releases conservative. *Response design.* The before/after comparison is biased towards ratios below one when a line is young. With 12 disclosures, a few weeks on each side and moves that cluster, the intervals are wide: a ratio near one rules out a large response, not a small one. *Vantage and liveness.* All data come from one node's view of the public graph; unannounced nodes, including most mobile wallets, are not covered, and liveness is inferred from peers' channel updates. *Fleets.* We identified one managed fleet by its alias and hubs. Smaller fleets without a shared alias would go undetected and be counted as independent nodes. *Exposure is not exploitability.* Several advisories require a channel with the victim, a particular configuration or inbound connections; we count nodes on affected releases, not nodes that can be attacked. Our advisory list is compiled from public sources and may miss silent fixes. *Association.* The hazard model describes which nodes upgrade sooner; it does not identify why.

X. ETHICS

Release-line inference identifies nodes running vulnerable software, and in 2026 Lightning deployments were exploited (BTCPay Server [40]) and CLN maintainers reported attacks on unpatched nodes [60]. We therefore publish aggregates only: no list, ranking, filter or label of individual nodes by release, platform or exposure, in the paper or in the released data. All inputs are public gossip and public release records, every advisory we study had been published before our analysis began, and the paper reports no new vulnerability.

XI. CONCLUSION

Lightning gossip carries no version string, but the feature bits each release announces identify its release line closely enough to measure who patches the network. Operators patch on release schedules: most upgrade within a few months of a release. Platforms patch on theirs: the largest single movements are node platforms shipping a release to all their users at once. Disclosures, which typically come long after the fix, move few nodes: of those we can test, only Core Lightning's of July 2026 raised the pace, and only among operators who were already close to current. A tail of small and neglected nodes stays on vulnerable releases. Measured by capacity it is small; measured by nodes, it depends on whether one counts a single fleet as hundreds of operators. For the maintainers of Lightning implementations, the lever that moves the tail is the platforms that package their releases.

DATA AND CODE AVAILABILITY

Code, the aggregate datasets behind every figure and table, and a script that recomputes every number quoted in the text are available at https://github.com/bitcoindatalabs/ln-upgrade-dynamics. All stages truncate their inputs at the data cutoff of 13 October 2026. The underlying gossip data are public; our daily gossip snapshots are available from the author on request.

REFERENCES

[1] Lightning Labs, "Security advisories for LND disclosed 2026-08-11 (ten advisories: three high, seven low severity)," https://security.lightning.engineering/, Aug. 2026, accessed 2026-10-07.

[2] ——, "Security advisories disclosed 2026-09-21 (four for lnd, four for btcd)," https://security.lightning.engineering/, Sep. 2026, accessed 2026-10-07.

[3] Core Lightning, "Core lightning has been triaging a high volume of AI-generated CVE reports … restart it with --offline," Post on X (@Core_LN), https://x.com/Core_LN/status/2092755509510283423, Aug. 2026, posted 2026-08-26 23:26 UTC. Accessed 2026-10-07.

[4] W. A. Arbaugh, W. L. Fithen, and J. McHugh, "Windows of vulnerability: A case study analysis," *IEEE Computer*, vol. 33, no. 12, pp. 52–59, 2000.

[5] E. Rescorla, "Security holes… who cares?" in *Proceedings of the 12th USENIX Security Symposium*. USENIX Association, 2003.

[6] S. Frei, T. Duebendorfer, and B. Plattner, "Firefox (in)security update dynamics exposed," *ACM SIGCOMM Computer Communication Review*, vol. 39, no. 1, pp. 16–22, 2009.

[7] Z. Durumeric, F. Li, J. Kasten, J. Amann, J. Beekman, M. Payer, N. Weaver, D. Adrian, V. Paxson, M. Bailey, and J. A. Halderman, "The matter of Heartbleed," in *Proceedings of the 2014 Internet Measurement Conference (IMC)*. ACM, 2014, pp. 475–488.

[8] S. K. Kim, Z. Ma, S. Murali, J. Mason, A. Miller, and M. Bailey, "Measuring Ethereum network peers," in *Proceedings of the Internet Measurement Conference (IMC 2018)*. ACM, 2018, pp. 91–104.

[9] P. Zabka, K.-T. Foerster, S. Schmid, and C. Decker, "Node classification and geographical analysis of the lightning cryptocurrency network," in *Proceedings of the 22nd International Conference on Distributed Computing and Networking (ICDCN)*. ACM, 2021, pp. 126–135.

[10] P. Espinasa-Vilarrasa, S. Sanvicente, C. Pérez-Solà, and J. Herrera-Joancomartí, "Decision tree based inference of lightning network client implementations," in *Modeling Decisions for Artificial Intelligence (MDAI 2024)*, ser. Lecture Notes in Computer Science, vol. 14986. Springer, 2024, pp. 103–114.

[11] S. Kumar, "Does client diversity protect the lightning network? Evidence from the 2026 Core Lightning shutdown," https://github.com/bitcoindatalabs/ln-client-diversity, 2026.

[12] Lightning Network Developers, "BOLT #7: P2P Node and Channel Discovery," https://github.com/lightning/bolts/blob/master/07-routing-gossip.md, accessed 2026-10-07.

[13] ——, "BOLT #9: Assigned Feature Flags," https://github.com/lightning/bolts/blob/master/09-features.md, accessed 2026-10-07.

[14] Umbrel, "Umbrel app store: commit history of `lightning/umbrel-app.yml` and `core-lightning/umbrel-app.yml`," https://github.com/getumbrel/umbrel-apps, accessed 2026-10-15.

[15] P. Zabka, K.-T. Foerster, S. Schmid, and C. Decker, "Empirical evaluation of nodes and channels of the lightning network," *Pervasive and Mobile Computing*, vol. 83, p. 101584, 2022.

[16] G. Kappos, H. Yousaf, A. Piotrowska, S. Kanjalkar, S. Delgado-Segura, A. Miller, and S. Meiklejohn, "An empirical analysis of privacy in the lightning network," in *Financial Cryptography and Data Security (FC 2021)*, ser. Lecture Notes in Computer Science. Springer, 2021, pp. 167–186.

[17] E. Rohrer, J. Malliaris, and F. Tschorsch, "Discharged payment channels: Quantifying the lightning network's resilience to topology-based attacks," in *2019 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*. IEEE, 2019, pp. 347–356.

[18] I. A. Seres, L. Gulyás, D. A. Nagy, and P. Burcsi, "Topological analysis of bitcoin's lightning network," in *Mathematical Research for Blockchain Economy (MARBLE 2019)*, ser. Springer Proceedings in Business and Economics. Springer, 2020, pp. 1–12.

[19] J.-H. Lin, K. Primicerio, T. Squartini, C. Decker, and C. J. Tessone, "Lightning network: a second path towards centralisation of the Bitcoin economy," *New Journal of Physics*, vol. 22, p. 083022, 2020.

[20] S. Martinazzi and A. Flori, "The evolving topology of the Lightning Network: Centralization, efficiency, robustness, synchronization, and anonymity," *PLOS ONE*, vol. 15, no. 1, p. e0225966, 2020.

[21] P. Zabka, K.-T. Foerster, C. Decker, and S. Schmid, "A centrality analysis of the Lightning Network," *Telecommunications Policy*, vol. 48, p. 102696, 2024.

[22] P. Casas, M. Romiti, P. Holzer, S. Ben Mariem, B. Donnet, and B. Haslhofer, "Where is the light(ning) in the taproot dawn? Unveiling the Bitcoin Lightning (IP) network," in *2021 IEEE 10th International Conference on Cloud Networking (CloudNet)*. IEEE, 2021, pp. 87–90.

[23] M. Romiti, F. Victor, P. Moreno-Sanchez, P. S. Nordholt, B. Haslhofer, and M. Maffei, "Cross-layer deanonymization methods in the lightning protocol," in *Financial Cryptography and Data Security (FC 2021)*, ser. Lecture Notes in Computer Science. Springer, 2021, pp. 187–204.

[24] F. Grötschla, L. Heimbach, S. Richner, and R. Wattenhofer, "On the lifecycle of a Lightning Network payment channel," in *Financial Cryptography and Data Security. FC 2025 International Workshops, Part I*, ser. Lecture Notes in Computer Science. Springer, 2025, pp. 1–16.

[25] N. Gögge, E. Rohrer, and F. Tschorsch, "On the routing convergence delay in the Lightning Network," in *Data Privacy Management, Cryptocurrencies and Blockchain Technology (DPM 2022, CBT 2022)*, ser. Lecture Notes in Computer Science, vol. 13619. Springer, 2023, pp. 203–218.

[26] D. Valko and J. Marx Gómez, "Geolocated Lightning Network topology snapshots: A dataset covering 2019–2023," *Scientific Data*, vol. 12, 2025.

[27] ——, "Topological and temporal stability analysis of the Lightning Network," arXiv:2512.20641, 2025.

[28] ——, "Shared channel capacity and node lifetime: An empirical study of the Lightning Network," arXiv:2606.20566, 2026.

[29] O. Osuntokun and C. Fromknecht, "Staying safe from the recent lightning vulnerability," Lightning Labs blog, Sep. 2019, https://blog.lightning.engineering/security/2019/09/27/cve-2019-12999.html.

[30] T. Neudecker, "Characterization of the Bitcoin peer-to-peer network (2015–2018)," Karlsruhe Institute of Technology, Karlsruhe Reports in Informatics 2019,1, 2019. [Online]. Available: https://publikationen.bibliothek.kit.edu/1000091933

[31] S. Park, S. Im, Y. Seol, and J. Paek, "Nodes in the Bitcoin network: Comparative measurement study and survey," *IEEE Access*, vol. 7, pp. 57 009–57 022, 2019.

[32] J.-P. Eisenbarth, T. Cholez, and O. Perrin, "A comprehensive study of the Bitcoin P2P network," in *2021 3rd Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS)*. IEEE, 2021, pp. 105–112.

[33] S. Zeisberg and V. Ulitzsch, "The blockchain ecosystem has a patch problem," Security Research Labs blog, May 2019, https://srlabs.de/blog/blockchain-patch-gap.

[34] E. Daniel, E. Rohrer, and F. Tschorsch, "Map-Z: Exposing the Zcash network in times of transition," in *2019 IEEE 44th Conference on Local Computer Networks (LCN)*. IEEE, 2019, pp. 84–92.

[35] A. Sarabi, Z. Zhu, C. Xiao, M. Liu, and T. Dumitraş, "Patch me if you can: A study on the effects of individual user behavior on the end-host vulnerability state," in *Passive and Active Measurement (PAM 2017)*, ser. Lecture Notes in Computer Science, vol. 10176. Springer, 2017, pp. 113–125.

[36] F. Li, Z. Durumeric, J. Czyz, M. Karami, M. Bailey, D. McCoy, S. Savage, and V. Paxson, "You've got vulnerability: Exploring effective vulnerability notifications," in *25th USENIX Security Symposium*. USENIX Association, 2016, pp. 1033–1050.

[37] A. Nappa, R. Johnson, L. Bilge, J. Caballero, and T. Dumitras, "The attack of the clones: A study of the impact of shared code on vulnerability patching," in *2015 IEEE Symposium on Security and Privacy (S&P)*. IEEE, 2015, pp. 692–708.

[38] F. Li and V. Paxson, "A large-scale empirical study of security patches," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2017, pp. 2201–2215.

[39] B. W. Turnbull, "The empirical distribution function with arbitrarily grouped, censored and truncated data," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 38, no. 3, pp. 290–295, 1976.

[40] BTCPay Server, "Security advisory: Update BTCPay Server to 2.4.2 immediately," https://blog.btcpayserver.org/security-advisory-btcpay-server-2-4-2/, Aug. 2026, published 2026-08-07. Accessed 2026-10-07.

[41] M. Morehouse, "DoS: Fake Lightning channels," Blog post, https://morehouse.dev/lightning/fake-channel-dos/, Aug. 2023, posted 2023-08-23. Accessed 2026-10-15.

[42] A. Riard, "Full disclosure: CVE-2023-40231 / CVE-2023-40232 / CVE-2023-40233 / CVE-2023-40234 "all your mempool are belong to us"," bitcoin-dev mailing list, https://mirror.b10c.me/lists/bitcoindev/CALZpt+GdyfDotdhrrVkjTALg5DbxJyiS8ruO2S7Ggmi9Ra5B9g@mail.gmail.com, Oct. 2023, posted 2023-10-16. Accessed 2026-10-15.

[43] M. Morehouse, "Invoice parsing bugs in CLN," Blog post, https://morehouse.dev/lightning/cln-invoice-parsing/, Dec. 2023, posted 2023-12-08. Accessed 2026-10-15.

[44] C. Pratap, "Vulnerability disclosure: Assertion DoS in Core Lightning," Delving Bitcoin, https://delvingbitcoin.org/t/vulnerability-disclosure-assertion-dos-in-core-lightning/2507, May 2026, posted 2026-05-16. Accessed 2026-10-15.

[45] ——, "Vulnerability disclosure: Twin memory exhaustion DoS vulnerabilities in Core Lightning," Delving Bitcoin, https://delvingbitcoin.org/t/vulnerability-disclosure-twin-memory-exhaustion-dos-vulnerabilities-in-core-lightning/2731, Jul. 2026, posted 2026-07-19. Accessed 2026-10-07.

[46] E. Cestari, "crashing a lightning node with a flood of pings," Blog post, https://erickcestari.dev/blog/ping-flood-oom/, Aug. 2026, posted 2026-08-25. Also posted to Delving Bitcoin as "Disclosure: Crashing CLN

with a flood of pings", https://delvingbitcoin.org/t/disclosure-crashing-cln-with-a-flood-of-pings/2846, 2026-08-27. Accessed 2026-10-15.
[47] N. Gögge, "Denial-of-service bugs in LND's channel update gossip handling," Delving Bitcoin, https://delvingbitcoin.org/t/denial-of-service-bugs-in-lnds-channel-update-gossip-handling/314, Dec. 2023, posted 2023-12-21. Accessed 2026-10-15.
[48] M. Morehouse, "DoS: LND onion bomb," Blog post, https://morehouse.dev/lightning/lnd-onion-bomb/, Jun. 2024, posted 2024-06-18. Accessed 2026-10-15.
[49] D. A. Harding, "Disclosure: irrevocable fees—stealing from LN using revoked commitment transactions," Delving Bitcoin, https://delvingbitcoin.org/t/disclosure-irrevocable-fees-stealing-from-ln-using-revoked-commitment-transactions/1314, Dec. 2024, posted 2024-12-11. Accessed 2026-10-15.
[50] M. Morehouse, "LND: Excessive failback exploit," Blog post, https://morehouse.dev/lightning/lnd-excessive-failback-exploit/, Mar. 2025, posted 2025-03-04. Accessed 2026-10-15.
[51] ——, "LND: gossip_timestamp_filter DoS," Blog post, https://morehouse.dev/lightning/lnd-gossip-timestamp-filter-dos/, Jul. 2025, posted 2025-07-22. Accessed 2026-10-15.
[52] ——, "LND: Replacement stalling attack," Blog post, https://morehouse.dev/lightning/lnd-replacement-stalling-attack/, Dec. 2025, posted 2025-12-04. Accessed 2026-10-15.
[53] ——, "LND: Excessive failback exploit #2," Blog post, https://morehouse.dev/lightning/lnd-excessive-failback-exploit-2/, Dec. 2025, posted 2025-12-04. Accessed 2026-10-15.
[54] ——, "LND: Infinite inbox DoS," Blog post, https://morehouse.dev/lightning/lnd-infinite-inbox-dos/, Dec. 2025, posted 2025-12-04. Accessed 2026-10-15.
[55] Lightning Labs, "LND gossip nil-map panic on zero-timestamp messages," Lightning Labs Security Advisories, https://security.lightning.engineering/2026/06/18/lnd-zero-timestamp-gossip-dos, Jun. 2026, posted 2026-06-18. Accessed 2026-10-15.
[56] ——, "LND HTLC first-stage sweep failure due to wallet budget constraint," Lightning Labs Security Advisories, https://security.lightning.engineering/2026/08/11/lnd-htlc-sweep-budget-failure, Aug. 2026, posted 2026-08-11. Accessed 2026-10-15.
[57] B. Teinturier, "Disclosure: LND doesn't wait for enough confirmations when closing channels," Delving Bitcoin, https://delvingbitcoin.org/t/disclosure-lnd-doesnt-wait-for-enough-confirmations-when-closing-channels/2800, Aug. 2026, posted 2026-08-13. Accessed 2026-10-15.
[58] Lightning Labs, "LND gossip stall via malformed channel_announcement flood," Lightning Labs Security Advisories, https://security.lightning.engineering/2026/09/21/lnd-gossip-validation-barrier-stall, Sep. 2026, posted 2026-09-21. Accessed 2026-10-15.
[59] ——, "LND out-of-memory via brontide write allocations," Lightning Labs Security Advisories, https://security.lightning.engineering/2026/09/21/lnd-brontide-write-allocation-oom, Sep. 2026, posted 2026-09-21. Accessed 2026-10-15.
[60] Core Lightning, "Urgent security update: If you're running version 26.06.7 or earlier, please upgrade to the latest release as soon as possible," Post on X (@Core_LN), https://x.com/Core_LN/status/2105835841847242976, Oct. 2026, posted 2026-10-02 01:42 UTC. Accessed 2026-10-15.